

Identifying Heart Disease in Patients Using Machine Learning Project

Avneet Bahra, Arinze Ezeifeka, Christopher A. Lee, and Joel Thomas

Fischell Department of Bioengineering, University of Maryland

ENBC 321: Machine Learning for Data Analysis

Professor Jamasp Jimmy Azarnoosh

April 27, 2026

Abstract

Cardiovascular disease kills more people worldwide than any other cause, making early detection and risk prediction a critical clinical challenge. For this project, we applied supervised and unsupervised machine learning methods to two datasets related to heart disease. The objective was to determine whether clinical and lifestyle variables could reliably predict the presence of heart disease or classify an individual's cardiovascular risk level. Dataset 1 framed heart disease prediction as a binary classification task, determining simply whether or not a patient has heart disease. It uses clinical features such as age, sex, chest pain type, cholesterol, resting blood pressure, maximum heart rate, and exercise-induced angina. Dataset 2 extended this to a multiclass problem, where the goal was to classify patients into distinct cardiovascular risk levels, rather than a simple yes/no outcome, based on demographic and lifestyle factors including weight, height, blood pressure, cholesterol, diabetes status, smoking habits, physical activity, and family history.

We trained and compared several supervised learning models such as logistic regression, support vector machines, k nearest neighbors, decision trees, random forests, and gradient boosting. To identify the best-performing approach, we looked at accuracy, precision, recall, F1 score, ROC AUC, confusion matrices, ROC curves, and precision recall curves. We also examined feature importance, and SHAP analysis to interpret model behavior and identify the key drivers of each prediction. For Dataset 1, gradient boosting and random forest achieved the strongest overall performance. For Dataset 2, performance was more mixed: random forest achieved the highest ROC AUC, logistic regression achieved the highest accuracy, and decision tree achieved the highest weighted F1 score. This indicated that no single model clearly outperformed all others across every metric. Some feature selection models exhibited unusually high accuracy, which was likely the result of label leakage as certain variables appeared to be directly derived from the target cardiovascular risk score. This highlights that while machine learning shows strong promise for pattern recognition in health data, careful preprocessing and leakage detection are critical to ensuring clinically valid results.

Introduction

Cardiovascular disease represents one of the most significant challenges in modern biomedicine, encompassing a broad range of conditions affecting the heart and blood vessels that can culminate in serious outcomes such as myocardial infarction, stroke, and heart failure.

Early detection is critical, as timely intervention through lifestyle modification, pharmacological treatment, and ongoing monitoring can meaningfully slow or prevent disease progression. In recent years, machine learning has emerged as a powerful tool in biomedical data analysis, capable of identifying complex, non-linear relationships among large sets of clinical variables that traditional statistical methods may overlook.

Prior studies have shown that machine learning models such as logistic regression, support vector machines, random forests, gradient boosting, and related ensemble methods can be effective for cardiovascular disease and heart disease prediction using clinical risk factors (Detrano et al., 1989; Pedregosa et al., 2011). These models are commonly evaluated using accuracy, ROC AUC, precision, recall, and F1 score, allowing researchers to compare both overall predictive performance and class-level behavior. Recent studies also highlight the importance of interpretability methods such as feature importance and SHAP, which help identify which clinical variables contribute most strongly to model predictions. This prior work supports the use of supervised learning methods for comparing model performance and identifying important cardiovascular risk variables in this project.

The central research question guiding this project was:

Can machine learning models effectively leverage clinical and lifestyle features to predict heart disease status or classify cardiovascular risk level?

To address this, we utilized two distinct datasets. The first was a binary classification dataset in which the target variable indicated the presence or absence of heart disease. The second was a cardiovascular risk dataset in which patients were grouped into one of three risk categories: LOW, INTERMEDIARY, and HIGH. These groupings were based on a combination of demographic, clinical, and lifestyle features.

This study pursued several connected objectives. First, we aimed to clean and preprocess the raw datasets to ensure data quality and suitability for modeling. From there, we applied a range of supervised machine learning algorithms covered throughout our course and systematically compared their performance using standard classification metrics. Beyond predictive accuracy, we aimed to interpret our models through feature importance, and SHAP analysis to identify which clinical and lifestyle variables carried the most predictive weight. Finally, we critically examined the limitations of our models, with particular attention to issues

such as label leakage and the broader question of whether our findings would generalize meaningfully to new, unseen patient populations.

Methodology

Dataset Description

Table 1. Demographic and clinical summary of the two cardiovascular datasets used in this study. Dataset 1 contained 1,025 patient records and was used for binary heart disease classification, while Dataset 2 contained 1,529 records and was used for multiclass cardiovascular risk classification. Dataset 2 included broader demographic, anthropometric, lifestyle, and clinical variables but also contained more missing values, requiring additional preprocessing before model training.

Characteristic	Dataset 1: Heart Disease Binary Classification	Dataset 2: Cardiovascular Risk Multiclass Classification
Sample Size	1,025	1,529
Number of Variables	14	22
Age, mean $\pm$ SD	54.43 $\pm$ 9.07	47.03 $\pm$ 12.42
Sex distribution	Male: 713 (69.6%); Female: 312 (30.4%)	Female: 773 (50.6%); Male: 756 (49.4%)
Target/class distribution	Disease: 526 (51.3%); No Disease: 499 (48.7%)	HIGH: 728 (47.6%); INTERMEDIARY: 581 (38.0%); LOW: 220 (14.4%)

Two cardiovascular-related datasets were used in this study. The first dataset was obtained from a Kaggle-hosted version of the UCI Heart Disease dataset, which originates from the Cleveland, Hungary, Switzerland, and VA Long Beach databases consisting of 1,025 patient records across 14 variables. The target variable was a binary indicator of heart disease presence, where a value of 0 denoted no heart disease and a value of 1 denoted a positive diagnosis. Predictor variables included age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, resting ECG results, maximum heart rate achieved, exercise-induced angina, ST depression, ST slope, number of major vessels, and thalassemia status.

The second dataset was obtained from the CAIR-CVD-2025 cardiovascular disease risk assessment dataset from Bangladesh, available through Mendeley Data was larger in scope,

comprising 1,529 patient records across 22 variables. The target variable, CVD Risk Level, was a multiclass outcome with three categories: LOW, INTERMEDIARY, and HIGH cardiovascular risk. Features in this dataset spanned a broader range of demographic, anthropometric, and clinical measures, including sex, age, weight, height, BMI, abdominal circumference, cholesterol, HDL, fasting blood sugar, estimated LDL, systolic and diastolic blood pressure, smoking status, diabetes status, physical activity level, family history of CVD, and a continuous CVD risk score.

Data Preprocessing

Dataset 1 required minimal preprocessing, as all features were numerical and no missing values were present. The dataset was split into training and testing sets using an 80/20 split, with stratification applied to preserve the proportional distribution of positive and negative heart disease cases across both subsets.

Dataset 2 required more extensive preprocessing because it included missing values, categorical variables, and several variables that were closely related to the target label. Missing numerical values were imputed using column-wise medians, while categorical values were handled using one-hot encoding after leakage-prone variables were removed. To reduce label leakage, Blood Pressure (mmHg), Blood Pressure Category, and CVD Risk Score were excluded from the modeling dataset. CVD Risk Score was especially problematic because it directly determined the CVD Risk Level target, meaning that retaining it would allow the model to reproduce the scoring rule rather than learn generalizable cardiovascular risk patterns. Removing these variables helped ensure that the reported model performance reflected prediction from selected clinical features rather than target-derived information.

A critical preprocessing step involved the removal of several variables to mitigate label leakage. Specifically, Blood Pressure (mmHg), Blood Pressure Category, and CVD Risk Score were excluded from the modeling dataset. CVD Risk Score is the direct determinant of the target variable, CVD Risk Level, meaning that retaining it would allow models to trivially replicate the scoring formula rather than learn generalizable biomedical relationships. Removing these features was essential to ensuring that model performance reflected genuine predictive capacity rather than information leakage from the target variable.

Machine Learning Models

Six supervised classification models were trained and evaluated across both datasets: logistic regression, support vector machine (SVM), k-nearest neighbors (KNN), decision tree, random forest, and gradient boosting. For models sensitive to feature scale, specifically logistic regression, SVM, and KNN, standardization was applied via a StandardScaler integrated directly into the modeling pipeline, ensuring that no data leakage occurred between the training and testing sets. Tree-based models, which are invariant to feature scaling, were trained without this preprocessing step.

Model Evaluation

Model performance for Dataset 1 was assessed using a comprehensive set of classification metrics, including accuracy, precision, recall, F1 score, and ROC AUC. Confusion matrices, ROC curves, and precision-recall curves were generated to provide a more complete picture of each model's predictive behavior across the decision threshold. For Dataset 2, the multiclass nature of the target variable required adaptations to the evaluation framework. Weighted precision, weighted recall, and weighted F1 score were computed to account for class imbalance across the LOW, INTERMEDIARY, and HIGH risk categories. ROC AUC was calculated using a one-vs-rest strategy, and multiclass confusion matrices were produced for each model to examine class-level misclassification patterns.

To interpret the actual internal decision-making of each model, several explainability techniques were applied. Feature importance scores were extracted from the random forest and gradient boosting models. Then used to assess the impact of each variable on model performance in a model-agnostic manner. SHAP summary plots were also generated to provide a unified, interpretable view of feature contributions across individual predictions.

Results

Dataset 1 Results

Dataset 1 exhibited a relatively balanced class distribution, comprising 526 heart disease cases and 499 non-disease cases. Exploratory correlation analysis revealed meaningful associations between the target variable and several clinical features, most notably chest pain type, maximum heart rate, exercise-induced angina, ST depression, number of major vessels, and thalassemia status, suggesting these variables would likely carry significant predictive weight in the modeling phase.

Supervised learning models performed strongly on the held-out test set, with tree-based models, particularly random forest and gradient boosting, achieving the highest overall performance. Confusion matrix analysis confirmed that the top-performing models correctly classified the large majority of both disease and non-disease cases, demonstrating strong discriminative ability across both classes.

Table 2. Dataset 1 model performance for binary heart disease classification. This table compares supervised learning models using accuracy, precision, recall, F1 score, and ROC AUC. Gradient boosting and random forest achieved the strongest overall performance, suggesting that tree-based ensemble models were most effective at distinguishing patients with and without heart disease.

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
Logistic Regression	0.8244	0.7949	0.8857	0.8378	0.8833
SVM	0.8732	0.8835	0.8667	0.8750	0.9591
k-NN	0.7854	0.7905	0.7905	0.7905	0.9336
Decision Tree	0.8585	0.9043	0.8095	0.8543	0.9400
Random Forest	0.9220	0.9083	0.9429	0.9252	0.9737
Gradient Boosting	0.9415	0.9266	0.9619	0.9439	0.9927

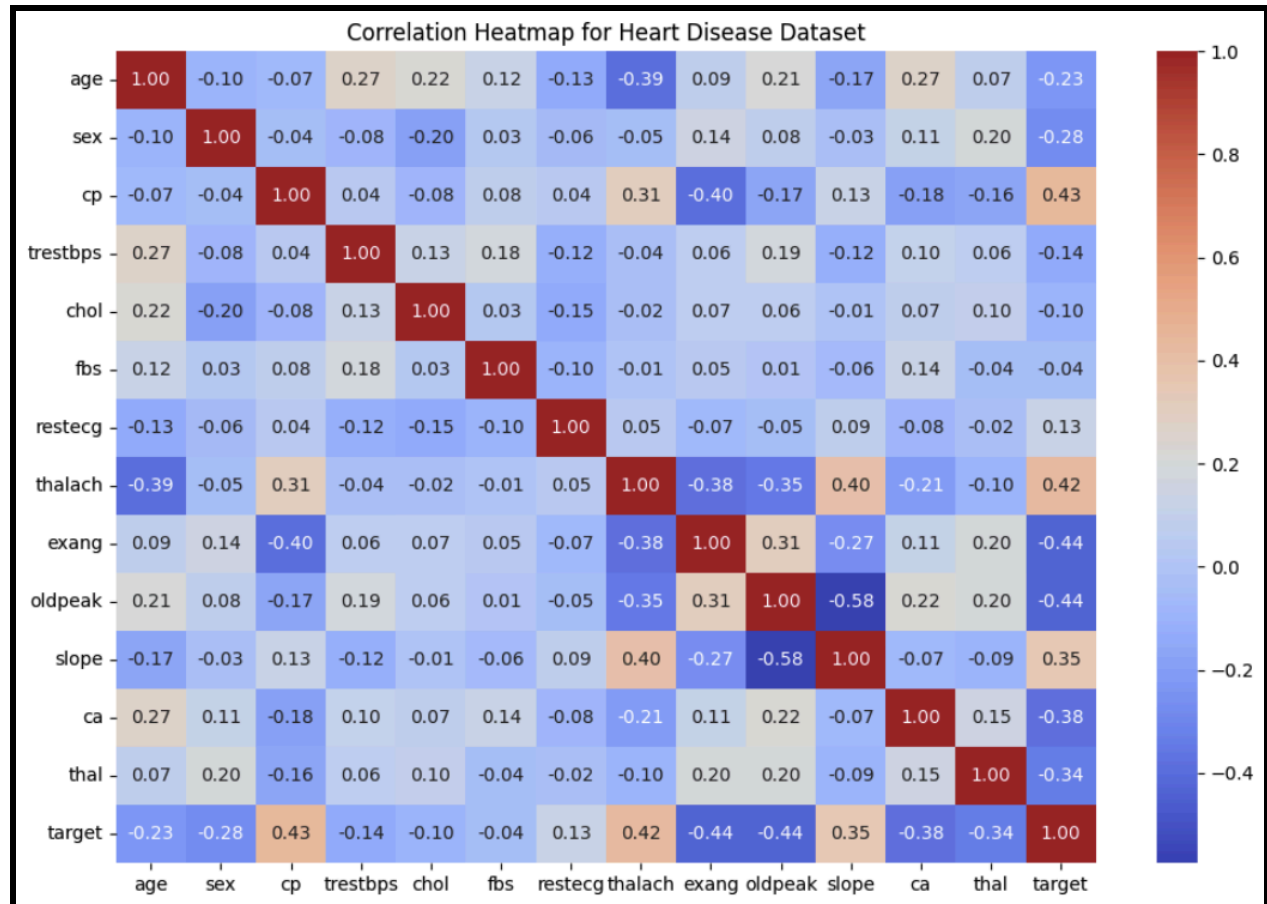


Figure 1. Correlation heatmap for Dataset 1 clinical variables. The heatmap shows pairwise correlations among the heart disease target and clinical predictors. Chest pain type, maximum heart rate, exercise-induced angina, ST depression, number of major vessels, and thalassemia showed stronger relationships with the target than many other variables, suggesting that these features may be important for model prediction.

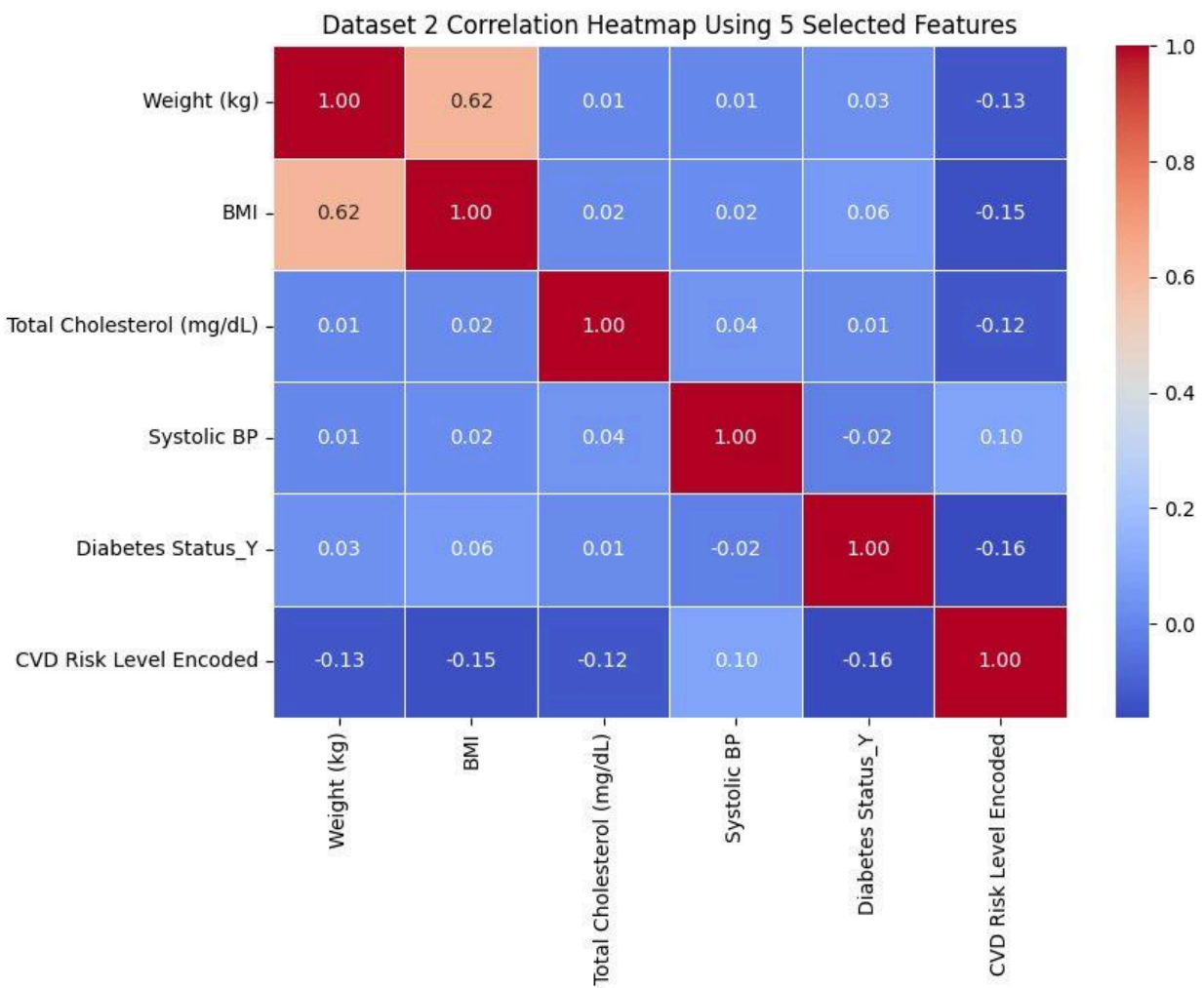


Figure 2. Correlation heatmap for Dataset 2 using the five selected predictor features and the encoded cardiovascular risk target. The heatmap shows pairwise Pearson correlations among weight, BMI, total cholesterol, systolic blood pressure, diabetes status, and the encoded CVD Risk Level target. CVD Risk Level Encoded was included only as a visualization reference to show how each selected feature relates to the LOW, INTERMEDIARY, and HIGH risk categories; it was not used as an additional predictor feature during model training. Overall, the selected predictors showed relatively weak linear correlations with the encoded target, which helps explain why Dataset 2 multiclass classification was more difficult than Dataset 1 binary classification.

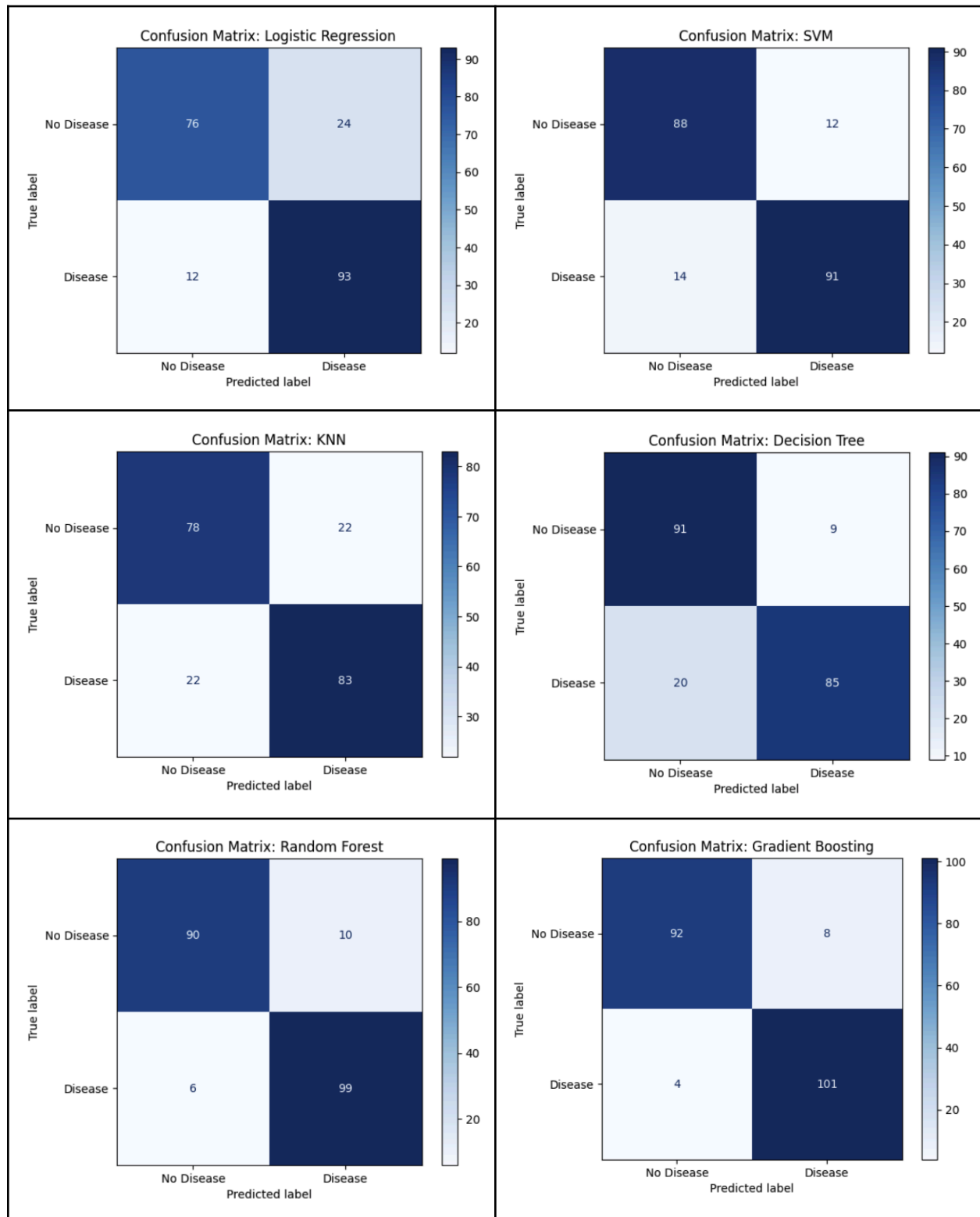


Figure 3. Confusion matrices for Dataset 1 supervised learning models. The confusion matrices compare predicted versus actual heart disease classifications for each model. Random forest and gradient boosting produced fewer

misclassifications than the simpler baseline models, supporting their stronger performance in distinguishing disease from non-disease cases.

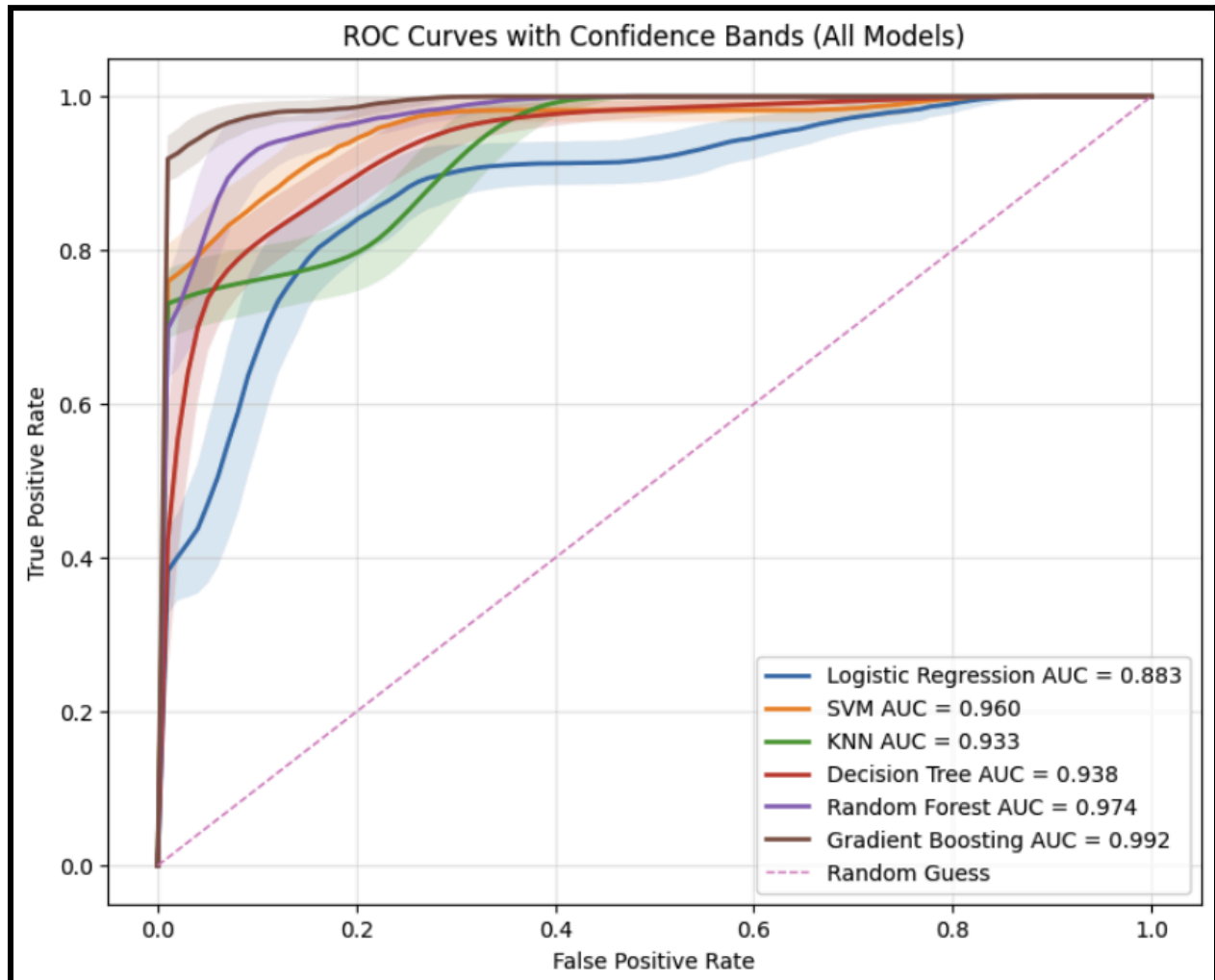


Figure 4. ROC curves for Dataset 1 models. The ROC curves show each model's ability to separate heart disease and non-disease cases across classification thresholds. Gradient boosting and random forest achieved the largest areas under the curve, indicating stronger overall discrimination compared with the other models.

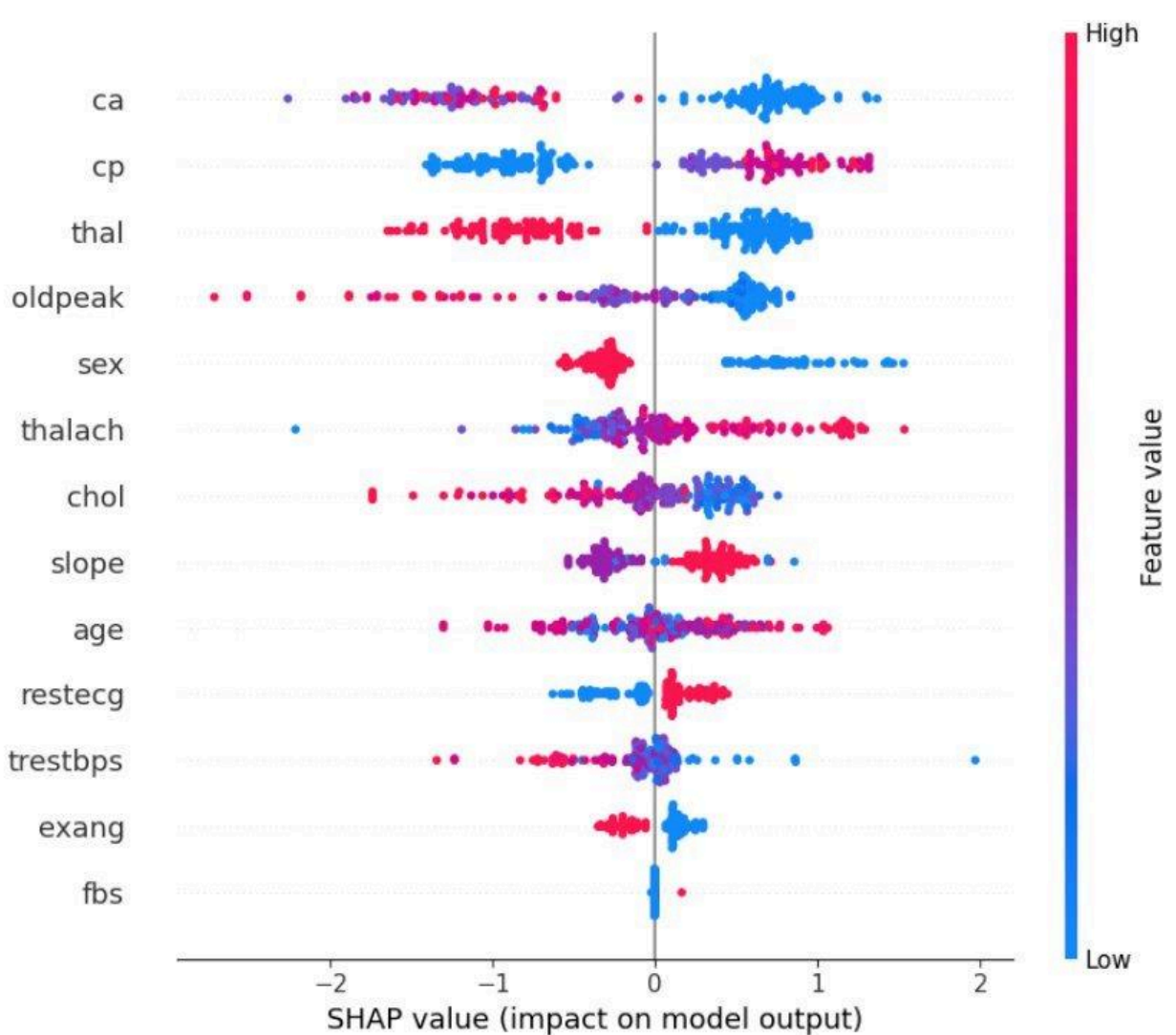


Figure 5. Feature importance analysis identified chest pain type, maximum heart rate, ST depression, exercise-induced angina, thalassemia status, and number of major vessels as the most influential predictors. These findings align with established biomedical understanding, as each variable directly reflects cardiac function or the heart's response to physiological stress. The plot's x-axis displays the SHAP value, indicating the impact on the model's output, while the color gradient reflects the actual feature value ranging from low to high.

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Dataset 2 Results

Dataset 2 contained three cardiovascular risk categories: 728 HIGH, 581 INTERMEDIARY, and 220 LOW risk cases. Because the LOW-risk group was much smaller than the other two classes, model performance was interpreted using weighted metrics in addition to accuracy and one-vs-rest ROC AUC. Compared with Dataset 1, Dataset 2 showed more moderate and mixed performance. Logistic regression achieved the highest overall accuracy at 0.581, random forest achieved the highest ROC AUC at 0.697, and decision tree achieved the highest weighted F1 score at 0.548. Therefore, no single model clearly dominated across all evaluation metrics. Random forest appeared strongest for overall class separation, while logistic regression and decision tree performed better on specific summary metrics. This suggests that multiclass cardiovascular risk stratification was more challenging than binary heart disease detection, likely due to weaker feature separation and class imbalance. .

Table 3. Dataset 2 model performance for multiclass cardiovascular risk classification. This table compares model performance for classifying patients into LOW, INTERMEDIARY, and HIGH cardiovascular risk groups using accuracy, weighted precision, weighted recall, weighted F1 score, and one-vs-rest ROC AUC. Logistic regression achieved the highest accuracy, random forest achieved the highest ROC AUC, and decision tree achieved the highest weighted F1 score. Because the best model varied by metric, Dataset 2 performance was considered mixed rather than being dominated by one classifier.

Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1 Score	ROC AUC
Random Forest	0.552	0.480	0.552	0.513	0.697
SVM	0.578	0.494	0.578	0.533	0.662
Gradient Boosting	0.519	0.483	0.519	0.492	0.653
Logistic Regression	0.581	0.495	0.581	0.533	0.696
Decision Tree	0.575	0.566	0.575	0.548	0.640
KNN	0.500	0.452	0.500	0.469	0.585

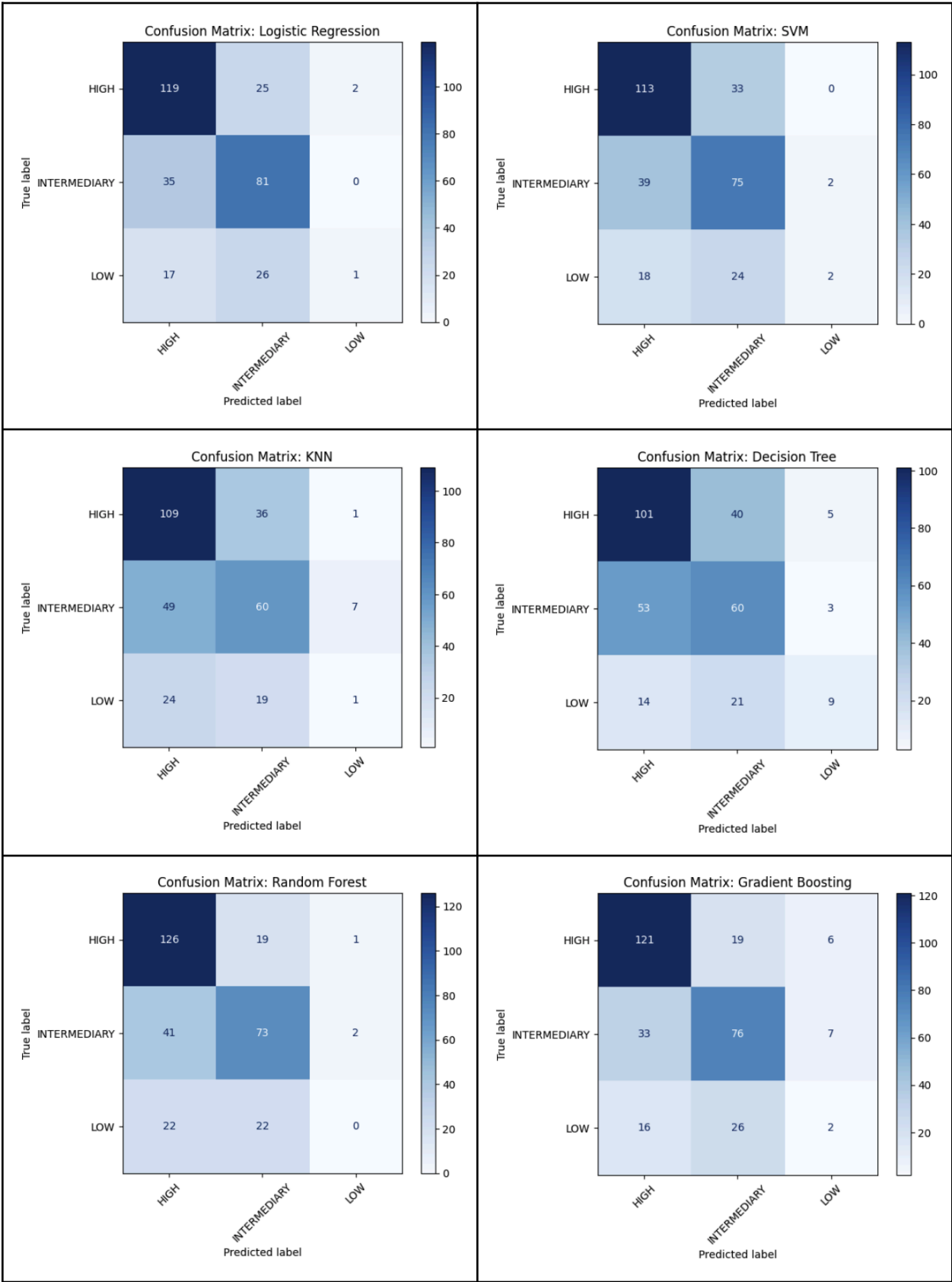


Figure 6. Confusion matrices for Dataset 2 multiclass cardiovascular risk models. The matrices show how well each model classified patients into LOW, INTERMEDIARY, and HIGH cardiovascular risk groups. Most models performed better on the HIGH and INTERMEDIARY groups than on the LOW group, likely because the LOW class had fewer samples and was underrepresented in the dataset.

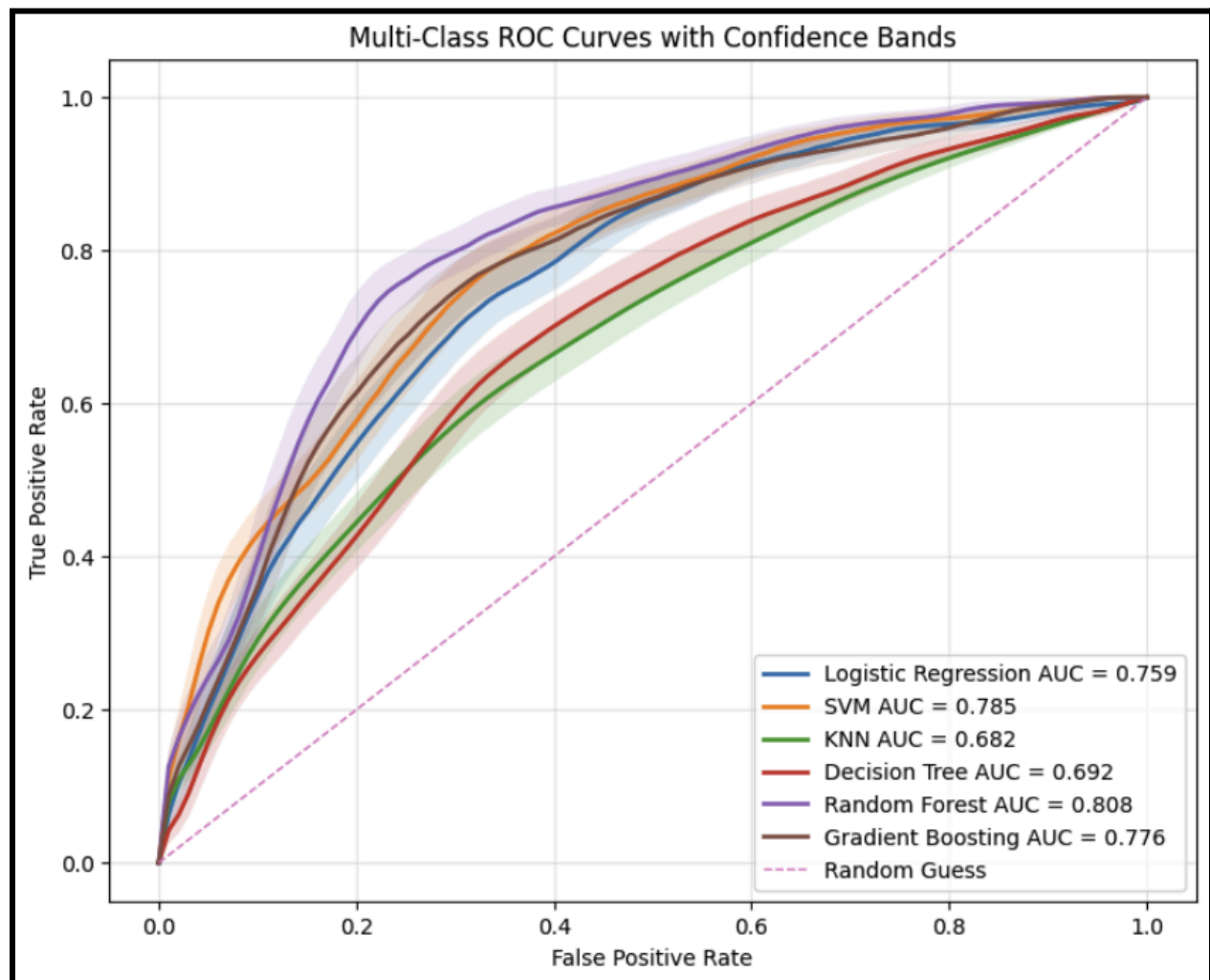


Figure 7. Multiclass ROC curves for Dataset 2 models using a one-vs-rest approach. The ROC curves summarize how well each model separated each cardiovascular risk class from the remaining classes. Random forest achieved the highest ROC AUC, suggesting the strongest overall class-separation ability among the models tested.

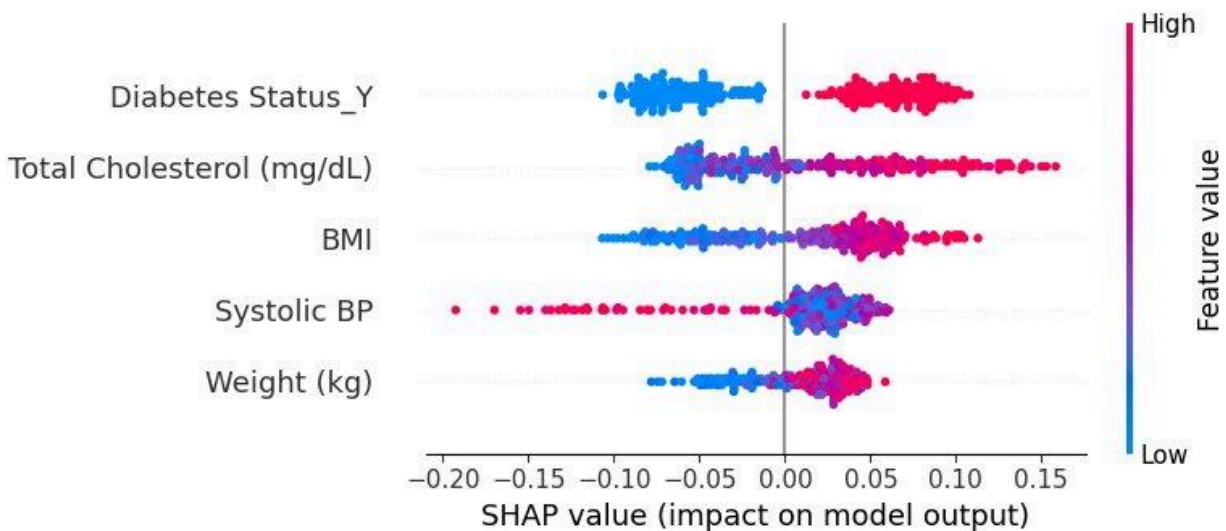


Figure 8. SHAP summary plot illustrating feature contributions for the multiclass cardiovascular risk models. This visualization demonstrates the impact of key lifestyle and clinical variables—including diabetes status, total cholesterol, BMI, systolic blood pressure, and weight—on the model's internal decision-making. The x-axis represents the SHAP value, which measures the effect on the model output, and the color scale maps the distribution of feature values for individual patients from low to high.

Discussion

The results of this study demonstrate that machine learning can serve as an effective tool for predicting heart disease status and cardiovascular risk level from clinical and lifestyle data. Dataset 1 yielded strong binary classification performance, particularly among tree-based models, confirming that meaningful predictive relationships exist between heart disease status and features such as chest pain type, maximum heart rate, exercise-induced angina, ST depression, number of major vessels, and thalassemia status. Gradient boosting achieved the strongest Dataset 1 performance overall, with 0.9415 accuracy and 0.9927 ROC AUC, while random forest also performed strongly with 0.9220 accuracy and 0.9737 ROC AUC. These results suggest that the binary heart disease dataset contained clearer feature-target relationships than the multiclass cardiovascular risk dataset.

Dataset 2 produced more moderate and mixed results rather than similarly strong results. Because Dataset 2 involved three risk categories, LOW, INTERMEDIARY, and HIGH, and had an imbalanced class distribution, performance was evaluated using weighted precision, weighted recall, weighted F1 score, and one-vs-rest ROC AUC. No single model clearly

dominated across all metrics. Logistic regression achieved the highest overall accuracy at 0.581, random forest achieved the highest ROC AUC at 0.697, and decision tree achieved the highest weighted F1 score at 0.548. This suggests that random forest was strongest for overall class separation, while logistic regression and decision tree performed better on specific summary metrics. Overall, Dataset 2 was more difficult to classify than Dataset 1, likely because the selected features showed weaker separation between neighboring risk groups.

Dataset 2 also required careful interpretation because several variables in the original dataset were directly or indirectly related to the cardiovascular risk scoring system. In particular, CVD Risk Score was leakage-prone because it was directly used to determine the target class, CVD Risk Level. Other derived variables, such as Blood Pressure Category and potentially estimated LDL, BMI, and waist-to-height ratio, may also encode parts of the underlying risk calculation. To reduce this issue, the final Dataset 2 modeling workflow removed the most leakage-prone variables and restricted the model input to five selected predictors: weight, BMI, total cholesterol, systolic blood pressure, and diabetes status. After this correction, the more realistic Dataset 2 performance, with the best accuracy around 58% and the best ROC AUC around 0.70, suggests that these selected clinical features contain useful predictive signal but are not sufficient to perfectly separate LOW, INTERMEDIARY, and HIGH risk groups.

The consistent strength of tree-based models in Dataset 1 reflects their ability to capture nonlinear relationships in clinical data. Cardiovascular risk does not necessarily scale linearly with variables such as age, cholesterol, blood pressure, or BMI, so ensemble methods like random forest and gradient boosting can model complex interactions more effectively than simpler linear approaches. However, this trend was less clear in Dataset 2, where logistic regression, random forest, and decision tree each performed best on different metrics. This reinforces that model selection should depend on the evaluation goal: accuracy, class balance, ROC AUC, or interpretability.

Several limitations should be addressed in future work. Both datasets were obtained from public repositories and may not fully represent real-world hospital populations. Dataset 1 was relatively small for clinical machine learning, with 1,025 samples, and Dataset 2 had class imbalance, especially in the LOW-risk group. In addition, the presence of derived variables in Dataset 2 created potential label leakage, which required careful feature removal before final interpretation. Finally, model evaluation was performed using held-out test sets from the same datasets rather than independent external patient cohorts. Future work should validate these

models on larger and more diverse clinical datasets, include deeper patient information such as medication use, longitudinal follow-up, imaging data, and detailed lifestyle history, and evaluate whether model performance differs across demographic groups such as age and sex.

Conclusion

Overall, this project demonstrated that machine learning can effectively predict heart disease presence and classify cardiovascular risk using demographic and clinical data. For Dataset 1, random forest and gradient boosting achieved strong binary classification performance, accurately distinguishing between disease and non-disease cases. For Dataset 2, multiclass models successfully stratified patients into LOW, INTERMEDIARY, and HIGH risk categories, provided that leakage-prone variables were rigorously excluded from the feature set.

Across both datasets, the most consistently influential predictors included blood pressure, cholesterol, BMI, heart rate, exercise response, chest pain type, diabetes status, smoking status, and family history of cardiovascular disease. Notably, these features align closely with the established clinical risk factors recognized in cardiovascular medicine, lending biological credibility to the patterns identified by our models.

Limitations

While this study yielded valuable predictive models, several limitations must be acknowledged. Primarily, Dataset 1 utilized a relatively small sample size of only 1025 patient records, which restricts the broader generalizability of the findings. For Dataset 2, the models exhibited weaker class separation. The multiclass algorithms frequently confused neighboring risk groups, struggling particularly with accurately isolating intermediate-risk patients. Furthermore, the analysis was constrained by feature limitations; highly relevant clinical factors such as medication use, family history, detailed lifestyle habits, and longitudinal follow-up data were not fully represented in the available data.

Future Work

There are several promising avenues for future research to build upon these findings. First, utilizing larger and more diverse datasets is essential; expanding the sample size and incorporating patients from varied demographic backgrounds will significantly improve model

reliability. To address the challenges observed in Dataset 2, future work must focus on improving multiclass risk prediction. This can be achieved by testing stronger hyperparameter tuning strategies, implementing class balancing techniques, and integrating additional clinical features to better differentiate between risk tiers.

Additionally, models should undergo rigorous external validation by testing them against independent CVD datasets to thoroughly evaluate their real-world generalizability. Finally, because CVD presentation and risk patterns vary—such as women frequently experiencing atypical signs rather than traditional chest pain—future models should explore sex-specific and demographic-aware analyses to determine if predictive performance varies across distinct patient subgroups.

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Appendix

- **Appendix A:** Python code for Dataset 1 preprocessing, model training, evaluation, ROC curves, feature importance, and SHAP analysis.
- **Appendix B:** Python code for Dataset 2 preprocessing, multiclass classification, confusion matrices, ROC curves, and random forest feature importance..
- **Appendix C:** Supplementary figures including correlation heatmaps, confusion matrices, ROC curves, precision recall curves, feature importance plots, and SHAP summary plots.